

Hospital Management System (HMS) - Patient Feedback & Analytics Platform

Transfer of Information (TOI) & Technical Handover Document

1. Executive Summary & Purpose

The **Hospital Management System (HMS) Patient Feedback & Analytics Platform** is an enterprise-grade web application engineered to capture, aggregate, analyze, and act upon multi-departmental inpatient (IP) and outpatient (OP) patient experiences in real time.

Key Objectives

- **Patient Engagement:** Deliver an intuitive, bilingual (English & Tamil), mobile-responsive feedback collection flow.
- **Administrative Action & Compliance:** Provide hospital administrators and quality officers with a real-time analytics dashboard, dynamic question management, complaint investigation logging (corrective & preventive actions), Excel/CSV multi-section export, and print-ready A4 executive reports.
- **Dual-Platform Architecture:** Function as a standalone cloud application on **Vercel** with PostgreSQL database and Node.js Serverless Functions, while remaining fully compatible with local hospital intranet **XAMPP / PHP / MySQL / PostgreSQL** infrastructure.

2. Technology Stack

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript + Vite	Component-driven, type-safe UI architecture
Styling & Design	Tailwind CSS v4 + Lucide Icons	Responsive modern interface, theme tokens, print CSS
State & Drag-and-Drop	@dnd-kit/core, @dnd-kit/sortable	Dynamic drag-and-drop question ordering
Notifications	sonner (Toast)	Non-blocking user feedback alerts
Cloud Backend	Vercel Serverless Node.js Functions (api/*.js)	RESTful serverless microservices
Cloud Database	PostgreSQL (Neon / @vercel/postgres)	Relational storage for feedback, complaints, questions
Local Backend	PHP 7.4 / 8.x (backend/admin/*.php - PHP 7.4+ Compatible)	On-premise XAMPP integration proxy
Build & Asset Sync	Node.js sync-dist.js	Synchronizes compiled Vite bundles across distributions
Development Platform	Google Antigravity IDE	Engineered and refactored with Google DeepMind's agentic AI IDE

3. Directory Structure & Architecture

HMS_V6.6/	
api/	# Vercel Serverless API Functions
db.js	# Centralized PostgreSQL Connection Pool
get-hospitals.js	# Hospital Profile & Branding
get-patient.js	# Patient UHID Verification
get-questions.js	# Dynamic Question Registry
get-responses.js	# Feedback Submissions & Complaint Reviews
login-ajax.js	# Admin Authentication
save-office-use.js	# Office Investigation & Resolution Action API
save-questions.js	# Dynamic Question CRUD API
submit-feedback.js	# Feedback Submission Ingestion API
archive/	# Archived Test & Scratch Scripts
test_scripts/	# Historical test and diagnostic PHP/JS files
backend/	# Local PHP/XAMPP Integration
admin/	# Dashboard & Login PHP Proxies
config/	# Local Database Config
frontend/	# Production Built Assets (Synced from Vite)
frontend_source/	# React + TypeScript Source Code
src/	
app/	
components/	
admin/	# Modular Admin Subcomponents
AdminSidebar.tsx	
AdminHeader.tsx	
OfficeUseModal.tsx	
common/	# Shared UI Controls (Stars, Emojis, Toggles)
StarRating.tsx	
EmojiRating.tsx	
ToggleSwitch.tsx	
ThreeStateToggle.tsx	
SelectableCard.tsx	
AdminDashboard.tsx	# Admin Coordinator Component
HospitalSelection.tsx	# Hospital Picker
WelcomePage.tsx	# Welcome Screen
App.tsx	# Patient Feedback Wizard Coordinator
real_db_data.ts	# Seed & Fallback Data Definitions
services/	# API Client & Export Logic
api.service.ts	# Centralized HTTP Client
export.service.ts	# 3-Section Structured CSV/Excel Engine
index.ts	# Barrel Export
types/	# Shared TypeScript Type Definitions
admin.types.ts	
feedback.types.ts	
office-use.types.ts	
index.ts	
styles/	# CSS & Print Style Rules
public/	# Static Assets & Public Index
sync-dist.js	# Build-time multi-target asset sync script
vercel.json	# Vercel Routing & Deployment Configuration
TOI_DOCUMENT.md	# This Knowledge Transfer Document

4. Patient Feedback Wizard Flow

graph TD

```
A[Patient Lands on Welcome Page] --> B[Step 1: Patient Verification & UHID Lookup]
B --> C[Step 2: Department Star/Emoji Ratings]
C --> D[Step 3: Department Yes/No Questions]
D --> E[Step 4: Overall Remarks & Recommend Hospital]
E --> F[API: /api/submit-feedback]
F --> G[PostgreSQL Database]
G --> H[Step 5: Thank You Confirmation Screen]
```

Wizard Lifecycle

1. **Language Selection:** Patients can toggle between **English** and **தமிழ் (Tamil)** at any time. All question labels, hints, and error alerts dynamically update.
2. **UHID Auto-Lookup:** When a patient enters their UHID, `/api/get-patient` pre-populates patient details (Name, Visit Type, Mobile, Department).
3. **Department Ratings:** Multi-department rating cards rendered with star scales (1 to 5) or animated emojis.
4. **Yes/No Checkpoints:** Binary/trinary service inquiries (Cleanliness, Cost transparency, Referral willingness).
5. **Inactivity Timer:** Automatic timeout after 60 seconds of inactivity on the Thank You screen, returning to the welcome screen for the next patient.

5. Original Hospital Feedback Form Specification & Digital Mapping

The web platform is an exact digital transformation of the hospital's standardized 5-page physical feedback and quality audit questionnaire, complete with bilingual English and Tamil localization:

Page 1: Patient Record (நோயாளி தகவல்)

- **Header:** Hospital Logo, Hospital Name, Address, Contact Number & Email.
- **Patient Identifiers:**
 - UHID (பதிவு எண்)
 - Patient Name (பெயர்)
 - Age (வயது)
 - Gender (பாலினம்: Male / Female / Other)
 - Mobile No (கைபேசி எண்)
 - Email (மின்னஞ்சல் முகவரி)
 - Address & City (முகவரி & நகரம்)
- **Visit & Admission Records:**
 - OP No & Date (புறநோயாளி எண் & தேதி)
 - IP No & Date (உள்நோயாளி எண் & தேதி)
 - Date of Admission (அனுமதிக்கப்பட்ட தேதி)
 - Date of Discharge (மருத்துவமனையை விட்டு சென்ற தேதி)

Page 2: Why Choose Us (நீங்கள் மருத்துவமனையை தேர்ந்தெடுத்தற்கான காரணம் என்ன?)

- Self Opinion (உங்கள் உள்ளுணர்வு)
- Advertisement / News (விளம்பரம் / செய்தி)
- Friends / Relatives (நண்பர்கள் / உறவினர்கள்)
- Corporate (நிறுவனம்)
- Employee (பணியாட்கள்)
- Referral Doctor (பரிந்துரைக்கப்பட்ட மருத்துவர்)
- Others (மற்றவை)

Pages 2 & 3: Department Service Ratings (சேவை கருத்துக்கள்)

Rated on a 4/5 scale: **Excellent** (மிக நன்று) | **Good** (நன்று) | **Average** (சுமார்) | **Poor** (மோசம்)

6. **Responsiveness at Reception** (வரவேற்பறையில் கவனிப்பு)
7. **Admission Process** (உள்சேர்க்கை முறை)
8. **Billing Services** (பில்லிங் சேவைகள்)
9. **Doctor's Treatment** (மருத்துவரின் கவனிப்பு)
10. **Nursing Care** (செவிலியர் சேவை)
11. **Pharmacy Services** (மருந்தக சேவைகள்)
12. **X-ray and Scan** (எக்ஸ்-ரே மற்றும் ஸ்கேன்)
13. **Laboratory Services** (ஆய்வக சேவைகள்)
14. **Insurance Services** (காப்பீட்டு சேவைகள்)
15. **Food Services** (உணவு சேவைகள்)
16. **Physiotherapy** (பிசியோதெரபி)
17. **Blood Bank Services** (இரத்த வங்கி சேவைகள்)
18. **Overall Service** (ஒட்டுமொத்த சேவைகளின் மதிப்பு)

Page 3: Service & Hygiene Inquiries (கேள்விகள்)

19. **Cleanliness of Hospital Environment (Toilets / Other areas):**

- *Tamil:* மருத்துவமனையின் சுற்றுப்புற தூய்மை (கழிப்பறைகள் / மற்ற இடங்கள்)
- *Options:* Yes / ஆம் | No / இல்லை (with specify location text area).

20. **Estimated Cost Explained at Admission Counter?:**

- *Tamil:* நீங்கள் உள்நோயாளியாக சேரும்போது மதிப்பிட்டு சிகிச்சை கட்டணம் எவ்வளவு ஆகும் என்று கூறப்பட்டதா?
- *Options:* Yes / ஆம் | No / இல்லை.

21. **Would you refer Hospital to your family / friends?:**

- *Tamil:* மருத்துவமனையின் சேவையை உங்கள் குடும்பத்திற்கும், நண்பர்களுக்கும் பரிந்துரைப்பீர்களா?
- *Options:* Yes / ஆம் | No / இல்லை.

Page 4: Suggestions & Staff Appreciation (மேம்பாடுகள் & பாராட்டுக்கள்)

22. **Improvement Suggestions:**

- *Prompt:* "Please specify if you have any suggestions for further improving our service" (எங்களின் சேவையை சிறந்த முறையில் முன்னேற்றுவதற்கு உங்களின் கனிவான கருத்துக்களை பகிர்ந்து கொள்ளவும்).

23. **Staff Appreciation:**

- *Prompt:* "Would you like to appreciate an individual or any particular service?" (நீங்கள் தனிப்பட்ட நபரையோ அல்லது குறிப்பிட்ட சேவையையோ பாராட்ட விரும்பினால் துறையை குறிப்பிடவும்).
- *Fields:* Staff Name (பெயர்) & Department (துறை).
- **Patient Signature Confirmation** (நோயாளியின் கையொப்பம்).

Page 5: For Office Use Only (அலுவலக பயன்பாட்டிற்கு மட்டும்)

Dedicated quality assurance audit and resolution section for hospital administration:

- **Review of the Complaint** (புகார் மீதான ஆய்வு): Clinical or service investigation notes.
- **Date of Review** (ஆய்வு செய்யப்பட்ட தேதி): Official administrative audit timestamp.
- **Corrective Action** (சரிசெய்யும் நடவடிக்கை): Immediate resolution taken for the patient.
- **Preventive Action** (தடுப்பு நடவடிக்கை): Systemic/protocol changes to prevent recurrence.
- **Incharge Name** (பொறுப்பாளர் பெயர்): Designated officer / Medical Superintendent.

6. Administrative Dashboard & Operational Workflows

The Admin Dashboard provides 7 core modules:

graph LR

```
Admin[Admin User] --> Overview[1. Overview KPIs]
Admin --> Responses[2. Feedback Responses Table]
Admin --> Report[3. Feedback Report & Print Engine]
Admin --> Analytics[4. Department Analytics]
Admin --> Questions[5. Dynamic Question Builder]
Admin --> Depts[6. Department Configuration]
Admin --> Branding[7. Hospital Profile & Branding]
```

Key Workflows:

24. Office Use & Problem Resolution Workflow:

- For unresolved patient feedback or low ratings, admins click "**Resolve Problem**" or "**Edit Office Review**".
- Opens OfficeUseModal to log:
- **Review of Complaint**: Detailed explanation of patient issue.
- **Date of Review**: Official review timestamp.
- **Corrective Action Taken**: Immediate corrective measures.
- **Preventive Action**: Long-term policy adjustments.
- **Incharge Name**: Officer responsible for resolution.
- Saves to complaint_review table via /api/save-office-use.

25. Clean 3-Section Excel / CSV Export:

- Generates structured CSV with UTF-8 BOM (\uFEFF) containing:
- **Section 1**: Department Rating Questions Summary (5★ distribution, counts & percentages).
- **Section 2**: Department Yes/No Questions Breakdown.
- **Section 3**: Individual Patient Responses & Office Resolution Log.

26. 100% Full-Width A4 Print Engine:

- Zero sidebar / header interference in print preview.
- Automatically renders cards into a clean **2-column grid** across the full page width.
- Avoids awkward page breaks inside cards using break-inside: avoid.

7. Database Schema Reference (PostgreSQL)

1. hospitals

Stores registered hospital profiles and branding.

- id (VARCHAR PRIMARY KEY) - e.g. apollo-hospital
- name (VARCHAR)
- logo_url (TEXT)
- contact_number (VARCHAR)
- email (VARCHAR)
- address (TEXT)

2. feedback_submission / patient_feedback

Stores individual patient feedback entries.

- id (SERIAL PRIMARY KEY)
- hospital_id (VARCHAR)
- uhid (VARCHAR)
- patient_name (VARCHAR)
- visit_type (VARCHAR) - IP or OP
- department_name (VARCHAR)
- overall_rating (NUMERIC)
- would_recommend (BOOLEAN)
- suggestions (TEXT)
- ratings (JSONB) - Key-value map of { [question_id]: score }
- yes_no_answers (JSONB) - Key-value map of { [question_id]: boolean }
- created_at (TIMESTAMP)

3. complaint_review

Stores administrative investigations and corrective actions.

- id (SERIAL PRIMARY KEY)
- hospital_id (VARCHAR)
- uhid (VARCHAR)
- review_of_complaint (TEXT)
- date_of_review (DATE)
- corrective_action (TEXT)
- preventive_action (TEXT)
- incharge_name (VARCHAR)
- created_at / updated_at (TIMESTAMP)

4. questions

Dynamic question registry for ratings and binary checks.

- id (VARCHAR PRIMARY KEY)
- hospital_id (VARCHAR)
- label_en (TEXT)
- label_ta (TEXT)
- question_type (VARCHAR) - rating or yes_no
- rating_mode (VARCHAR) - star or emoji
- category (VARCHAR)
- order_index (INTEGER)
- is_deleted (BOOLEAN)

8. API Endpoints Reference

Endpoint	Method	Description
/api/get-hospitals	GET	Fetches hospital profile, branding, and config
/api/get-patient	GET	Validates UHID and returns patient registration data
/api/get-questions	GET	Returns list of active and past dynamic feedback questions
/api/get-responses	GET	Returns feedback submissions merged with complaint review actions
/api/submit-feedback	POST	Ingests new patient feedback submissions
/api/save-office-use	POST	Upserts complaint reviews, corrective, and preventive actions
/api/save-questions	POST	Saves custom questions, bilingual labels, and ordering
/api/login-ajax	POST	Validates admin credentials and issues session token

9. Complete Step-by-Step Setup Guide (Beginner Friendly)

A. Prerequisites Installation

27. Install **Node.js LTS (v18+)** from nodejs.org.
28. Install **Git** from git-scm.com.
29. Install **VS Code** (recommended editor).

B. Supabase Cloud Database Setup

30. Create a free account at supabase.com.
31. Create a new project (e.g. `hms-feedback-db`), set a database password, and choose your preferred region.
32. In Supabase, navigate to **SQL Editor** $\rightarrow$ **New query**.
33. Open `database_dump_with_data.sql`, copy the entire SQL script, paste into SQL Editor, and click **Run**.
34. Retrieve your database connection settings from **Project Settings** $\rightarrow$ **Database**:

- Host, Port (5432), Database (postgres), User (postgres.[REF]), Password.

C. Local Project Setup & Development

35. Clone the repository:

```
`bash
git clone https://github.com/Avaneesh-ravi/HMS_2.4.git
cd HMS_2.4
`
```

36. Navigate into `frontend_source/` and install dependencies:

```
`bash
cd frontend_source
npm install
`
```

37. Start the local development server:

```
`bash
npm run dev
`
```

Open `http://localhost:5173/` in your browser.

38. Build production bundles:

```
`bash
npm run build
`
```

(Compiles React components and automatically syncs to `frontend/`, `api/frontend/`, and `public/` via `sync-dist.js`).

D. Vercel Cloud Deployment Setup

39. Sign in to vercel.com with GitHub.
40. Click **Add New...** $\rightarrow$ **Project** and select `HMS_2.4`.
41. Configure Environment Variables in Vercel:

- `DB_HOST`: Your Supabase Host
- `DB_PORT`: 5432
- `DB_NAME`: postgres
- `DB_USER`: Your Supabase User
- `DB_PASS`: Your Supabase Password

42. Click **Deploy**. Vercel will build and publish your project to a public *.vercel.app URL.

E. Local XAMPP Setup (Intranet Mode)

43. Install XAMPP and place the project folder in c:\xampp\htdocs\HMS_V6.6.

44. Start **Apache** in XAMPP Control Panel.

45. Access Feedback Form: http://localhost/HMS_V6.6/frontend/feedback-form.php?hospital_id=apollo-hospital.

46. Access Admin Dashboard: http://localhost/HMS_V6.6/backend/admin/login.php.

10. Maintenance & Handover Checklist

- [x] **Zero Code Duplication**: Types extracted to src/types/, services extracted to src/services/.
- [x] **Root Cleanliness**: Scratch and test PHP files archived into archive/test_scripts/.
- [x] **Print Output Verification**: Verified 100% full-width layout without sidebar interference.
- [x] **CSV/Excel Export Verification**: Verified 3-section layout with UTF-8 BOM encoding.
- [x] **Complaint Investigation Logging**: Tested "Resolve Problem" and "Edit Office Review" workflows.
- x] ****QA Defect Log Verification****: All 27 historical test cases resolved and documented in [TEST_REPORTS_AND_CORRECTIONS.md`.

11. Development & Engineering Credit

This project architecture, modular refactoring, bilingual dynamic question engine, reporting system, and documentation were developed and optimized with the assistance of **Google Antigravity IDE** (Google DeepMind Advanced Agentic AI Coding Environment).